

Raymond Jiang

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EDUCATION

Baruch College (CUNY)

M.S. in Statistics – Data Science Track

B.B.A. in Statistics & Quantitative Modeling; Minor in Mathematics

New York, NY

Sep 2026 – May 2028

May 2026

EXPERIENCE

Columbia University Irving Medical Center

Machine Learning Researcher

New York, NY

Jun 2026 – Present

- Developed machine learning pipeline predicting HIV-1 antibody neutralization across 26 broadly neutralizing antibodies and 3 IC50 thresholds using ESM protein language-model representations and TabPFN classifier.
- Evaluated model scaling from 200 to 1,600 training sequences across 3 random seeds, measuring MCC, AUC, accuracy, and F1 to quantify training-set size impact on predictive performance.
- Reduced 1.8M embedding features to 1,000–2,000 selected features using SelectKBest and ANOVA F-statistics, eliminating 99.8% of original feature space before classification.
- Evaluated generalization on 396 independent HIV Env sequences and built reproducible inference pipelines comparing performance across antibodies, thresholds, training sizes, and random seeds.
- Automated GPU research workflows on Columbia's C2B2 HPC cluster using Python, Bash, SLURM, and CUDA for sequence alignment, embedding generation, training, inference, and experiment aggregation.

CUNY Tech Prep

Data Science Fellow

New York, NY

Jul 2025 – May 2026

- Selected for competitive year-long technical fellowship spanning 11 CUNY colleges, focused on machine learning, data science, software engineering, and technical interview preparation.
- Built end-to-end data science projects using Python, Pandas, NumPy, Scikit-learn, SQL, and Git while applying data cleaning, feature engineering, model evaluation, and visualization workflows.

PROJECTS

InterPrep – AI Interview Trainer | *Python, FastAPI, Gemini LLM, Whisper* | [Live App](#) Sep 2025 – Dec 2025

- Collaborated on full-stack voice AI interview platform generating structured 100-point interview evaluations across 8 algorithm categories.
- Engineered Pandas/NumPy data pipeline indexing 1,800+ LeetCode problems by topic, difficulty, and company tag for rapid question retrieval.
- Implemented faster-Whisper speech-to-text pipeline for audio ingestion, segmentation, and transcript generation, achieving approximately 90% transcription accuracy on tested technical vocabulary.
- Integrated FastAPI backend services with Streamlit interface to support interactive voice sessions and AI-generated feedback.

NYC Bus Lane Analysis – ACE & Congestion Pricing | *Python, Pandas, GeoPandas* | [GitHub](#) Jun 2025

- Analyzed NYC bus performance from 2020–2024 across 100+ routes to investigate changes associated with Automated Camera Enforcement (ACE).
- Processed and joined 1M+ enforcement records with geospatial route data using Pandas and GeoPandas to identify enforcement patterns by route and time.
- Identified approximately 13% higher speeds on the M101 and 12% on the M2 following ACE implementation in analyzed periods.

TECHNICAL SKILLS

Languages & Core: Python (Pandas, NumPy, Scikit-learn), SQL, C++, TypeScript, JavaScript

Machine Learning & AI: PyTorch, TensorFlow, TabPFN, ESM, Classification, Regression, Feature Selection

Tools & Infrastructure: FastAPI, Git, Linux, Bash, SLURM, CUDA